

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Sixth Semester B.E. (CSE) (Credit System) Degree Examinations
May - June 2023

20CSE21 - INTERNET OF THINGS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Define IoT and explain any four characteristics of IoT.	10	L*2	1	1
b)	Apply the IoT in the healthcare industry to improve patient care and reduce healthcare costs.	10	L3	1	1
2. a)	What is a sensor, and how is it used in IoT? Provide three examples of a sensor used in an IoT application.	10	L2	1	1
b)	Explain the IoT Level – 3 deployment template and design a concept of smart home application in Level – 3.	10	L3	1	1
3. a)	What are communication models? Explain any two communication models.	8	L2	1	1
b)	Develop a python program to read the data from the user and write to a text file.	4	L3	2	1
c)	Illustrate how python is used in IoT applications.	8	L2	2	1
Unit – II					
4. a)	Explain the steps involved in installing Linux on Raspberry Pi.	8	L2	5	1
b)	Develop a sketch and diagram for Arduino that will use an ultrasonic sensor and blink a LED when an obstacle is detected.	8	L3	3	1
c)	Develop a program to demonstrate the working of temperature sensor.	4	L3	3	2
5. a)	Develop a program to rotate a servo motor when a temperature of > 50 degrees centigrade is recorded.	10	L3	4	2
b)	What are the different models of Arduino boards? How are they different?	5	L2	4	1
c)	Demonstrate the usage of a push button with a suitable program.	5	L3	3	1
6. a)	Develop a python program and write the circuit diagram for Raspberry Pi to read the serial output from the Arduino.	10	L3	4	2
b)	Explain the board design of a Raspberry Pi 3B.	6	L2	5	1
c)	Explain the working principle of a potentiometer.	4	L2	3	1
Unit – III					
7. a)	Design a program in python to communicate between raspberry pi and two LEDs, making the LEDs to blink alternatively.	10	L3	5	2
b)	List the differences between Arduino and Raspberry Pi.	10	L1	3	1
8. a)	Develop a python program to read the data from the DHT22 sensor using a Raspberry pi. Store the data on a cloud database. Generate a beep sound when the temperature crosses a threshold.	10	L3	5	2
b)	Design a python program to read the value from an ultrasonic sensor and generate a sound when an obstacle is detected.	10	L3	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
